A resource obstruction for qubit measurement self-testing with matched kernel preparations

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Abstract

Resource counts in dimension-bounded self-testing depend on both the unknown measurement and the reference preparations. We consider a complex-qubit prepare-and-measure experiment with exactly four preparations, binary auxiliary settings, and an extremal four-outcome target positive operator-valued measure. The reference preparation associated with each target outcome is its pure kernel state. We prove that every real linear witness on the complete probability table that is globally maximized at this reference admits an inequivalent maximizing target with at most three nonzero effects whenever the auxiliary observable Bloch vectors span at most two dimensions. The proof first uses measurement optimization and complementary slackness to force all target coefficients into a common-row-shift plus nonnegative exclusion-penalty form. Preparation optimality then leaves at most two inputs with nonzero auxiliary contributions, allowing an explicit positive and complete replacement measurement. The result includes biased and degenerate binary measurements and allows input-independent shared randomness. It obstructs reducing auxiliary resources within the matched-kernel reference class used by existing constructions. It does not establish unrestricted resource minimality, a bound on preparation number, or robustness to noise.

Keywords: quantum self-testing; qubit measurements; prepare-and-measure scenarios; resource bounds; semidefinite duality.

1 Introduction

Dimension-bounded prepare-and-measure self-testing certifies an unknown quantum realization from observed probabilities and a bound on the transmitted Hilbert space. For a nonprojective qubit measurement, the number of preparation inputs and auxiliary measurement settings is an experimentally relevant resource. Tavakoli *et al.* [1] developed a construction that first certifies suitable preparations and then adds penalties for target outcomes excluded by those states. Drótos *et al.* [2] obtained four-preparation, three-binary-auxiliary constructions for arbitrary extremal four-outcome qubit measurements and conjectured their resource minimality. Their preparation-only optimization places optimal state directions in the auxiliary Bloch span. By itself, that observation does not constrain a general witness whose target probabilities also enter preparation optimization.

Here we address this distinction within the matched state-exclusion reference class. All sixteen target coefficients are initially arbitrary, and the witness need only certify the target. Measurement optimality nevertheless forces the target contribution to be a constant plus nonnegative exclusion penalties. We then construct an inequivalent target attaining the same

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global score when the auxiliary Bloch span has dimension at most two. Thus changing the weights of the complete probability table cannot reduce the auxiliary count at these references.

This restriction is the one used in the cited constructions, rather than an assumption about all possible self-tests. The revised semi-SIC construction [3] also uses matched exclusion states; its discussion asks about smaller setups. General preparation and measurement self-testing constructions [4] establish existence with other auxiliary resources. A recent photonic semi-SIC realization [5] uses four preparations, three dichotomic settings and the target. Our result supplies a necessary resource condition for matched references; the unrestricted minimality question is not resolved here.

2 Model and reference strategies

Alice receives an input $i \in \{1,2,3,4\}$ and sends a single complex qubit to Bob. Bob chooses either a binary auxiliary setting $y \in \{1,\dots,K\}$ or the target setting t, which has four labeled outcomes. A deterministic strategy consists of density operators $\rho_i \geq 0$, $\operatorname{Tr}\rho_i = 1$, binary effects $B_{b|y} \geq 0$, $B_{0|y} + B_{1|y} = \mathbb{I}$, and target effects $E_j \geq 0$, $\sum_{j=1}^{4} E_j = \mathbb{I}$. The probabilities are

$$p(b|i,y) = \operatorname{Tr}(\rho_i B_{b|y}), \qquad p(j|i,t) = \operatorname{Tr}(\rho_i E_j). \tag{1}$$

The competing target may have zero effects or be nonextremal. No entanglement assistance, trusted axes, sequential instrument or additional communication is allowed. Input-independent shared randomness permits convex mixtures of these strategies. All quantities are dimensionless, and the problem is static and exactly noiseless.

Use the Pauli convention $(\sigma_x, \sigma_y, \sigma_z)$, with $(\sigma_y)_{01} = -\mathrm{i}$. The reference is

$$\rho_i^\star = \frac{\mathbb{I} + \mathbf{r}_i \cdot \boldsymbol{\sigma}}{2}, \qquad T_i = \alpha_i(\mathbb{I} - \mathbf{r}_i \cdot \boldsymbol{\sigma}), \tag{2}$$

where

$$\alpha_i > 0, \quad \|\mathbf{r}_i\| = 1, \quad \sum_i \alpha_i = 1, \quad \sum_i \alpha_i \mathbf{r}_i = 0. \tag{3}$$

The four T_i are linearly independent Hermitian matrices, as required for an extremal four-outcome qubit measurement. In particular, $\operatorname{Tr}(\rho_i^\star T_i) = 0$: each preparation is the pure state supported on the kernel of its matched effect. For the reference binary observables write

$$O_y := B_{0|y} - B_{1|y} = \beta_y \mathbb{I} + \mathbf{d}_y \cdot \boldsymbol{\sigma}, \qquad |\beta_y| + \|\mathbf{d}_y\| \leq 1. \tag{4}$$

The inequality is equivalent to positivity of both binary effects; it includes projective, biased, unsharp and degenerate measurements.

Binary normalization puts every real linear witness on the complete table into the form

$$L = C_0 + \sum_{i,y} w_{iy} \operatorname{Tr}(\rho_i O_y) + \sum_{i,j} c_{ij} \operatorname{Tr}(\rho_i E_j), \tag{5}$$

with arbitrary real w_{iy}, c_{ij}. We say that L self-tests the reference target if every globally maximizing admissible realization has target effects related to T_i by one common unitary or antiunitary transformation. The input and outcome labels are fixed. The obstruction below would remain valid even if target outcome permutations were allowed, since it changes the number of nonzero effects. No separate preparation self-test is assumed.

3 Supporting witnesses and the resource obstruction

Proposition 1. *Let the reference satisfy Eqs.* (2)–(4)*, with four linearly independent target effects. Suppose* $\dim \operatorname{span}\{\mathbf{d}_y\}_{y=1}^{K} \le 2$*. If a real linear witness on the complete probability table attains its global maximum at this reference, it also has an admissible maximizing realization with the same auxiliary measurements and a target with at most three nonzero effects. Consequently it cannot self-test the reference four-outcome target. This includes any two arbitrary binary auxiliary measurements and allows input-independent shared randomness.*

We prove the proposition through a necessary coefficient form and an explicit replacement strategy.

3.1 A basis property of the reference

The four $\rho_i^\star$ form a basis of the real vector space of Hermitian 2×2 matrices. Indeed, independence of the T_i is independence of the coefficient vectors $(1, -\mathbf{r}_i)$, which is unchanged by reversing the three Pauli coordinates. Furthermore, every triple of $\mathbf{r}_i$ is linearly independent. If one triple spanned at most a plane through the origin, the positive-weight relation in Eq. (3) would place the fourth direction in the same plane. All four effects would then belong to a Hermitian subspace of dimension at most three, a contradiction. Thus no plane through the origin contains more than two reference directions, and no two are collinear.

3.2 The target coefficient normal form

Lemma 1. *If the reference maximizes L, there exist real numbers* ℓ_i *and nonnegative numbers* k_j *such that*

$$c_{ij} = \ell_i - k_j \delta_{ij}. \tag{6}$$

For every admissible strategy, its target contribution is therefore

$$\sum_{i,j} c_{ij} p(j|i,t) = \sum_i \ell_i - \sum_i k_i p(i|i,t). \tag{7}$$

Proof. Hold the reference preparations and auxiliaries fixed. Global optimality implies that the reference target solves the measurement semidefinite program

$$\begin{aligned} &\text{maximize} \quad \sum_j \operatorname{Tr}(H_j E_j), \qquad H_j = \sum_i c_{ij} \rho_i^\star, \\ &\text{subject to} \quad E_j \ge 0, \qquad \sum_j E_j = \mathbb{I}. \end{aligned} \tag{8}$$

Its dual minimizes $\operatorname{Tr}\Lambda$ over Hermitian Λ with $\Lambda - H_j \ge 0$ for each j. This follows directly from the Lagrangian $\operatorname{Tr}\Lambda + \sum_j \operatorname{Tr}[(H_j - \Lambda)E_j]$. The primal is strictly feasible at $E_j = \mathbb{I}/4$ and has a compact feasible set. The dual is strictly feasible at $\Lambda = t\mathbb{I}$ for sufficiently large t. Finite-dimensional Slater duality [6] therefore provides an attained dual optimum and complementary slackness:

$$S_j := \Lambda - H_j \ge 0, \qquad \operatorname{Tr}(S_j T_j) = 0. \tag{9}$$

For positive operators, zero trace of the product implies orthogonal supports. Since T_j has rank one on a two-dimensional space, its complementary support is the one-dimensional support of $\rho_j^\star$. Hence

$$S_j = k_j \rho_j^\star, \qquad k_j \ge 0. \tag{10}$$

This includes zero slack. Expand the common dual operator uniquely as $\Lambda = \sum_i \ell_i \rho_i^\star$. Comparing coefficients in the state basis gives Eq. (6).

This equality constrains the numerical coefficients of the original witness, rather than merely its value at the reference. It consequently remains true when preparations and measurements change. Summing each target probability row to one gives Eq. (7). □

3.3 Only two preparations can contribute to auxiliary optimization

Keep all reference measurements fixed and define

$$\mathbf{u}_i = \sum_y w_{iy} \mathbf{d}_y. \tag{11}$$

For a varied preparation $\rho_i = (\mathbb{I} + \mathbf{s} \cdot \boldsymbol{\sigma})/2$, Eq. (7) makes the state-dependent witness term

$$(\mathbf{u}_i + k_i \alpha_i \mathbf{r}_i) \cdot \mathbf{s}, \qquad \|\mathbf{s}\| \le 1. \tag{12}$$

Because $\mathbf{r}_i$ is globally optimal with these measurements fixed,

$$\mathbf{u}_i + k_i \alpha_i \mathbf{r}_i = \tau_i \mathbf{r}_i, \qquad \tau_i \ge 0. \tag{13}$$

The case $\tau_i = 0$ allows a completely vanishing preparation gradient. In particular $\mathbf{u}_i = (\tau_i - k_i \alpha_i)\mathbf{r}_i$; its multiplier may have either sign. Every nonzero $\mathbf{u}_i$ is parallel to a reference direction and belongs to the auxiliary span. The basis property above and the span hypothesis imply

$$A := \{i : \mathbf{u}_i \neq 0\}, \qquad |A| \le 2. \tag{14}$$

For an inactive label, the auxiliary contribution is independent of the preparation. Individual w_{iy} need not vanish: dependent or degenerate auxiliary directions can cancel. The constants $\sum_y w_{iy} \beta_y$ are also preserved if the auxiliaries are unchanged.

3.4 A positive complete target with the same score

Suppose first that $A = \{a, b\}$. Let c, d be the two inactive labels, and set

$$s = \|\mathbf{r}_a + \mathbf{r}_b\|, \qquad \mathbf{q} = \frac{\mathbf{r}_a + \mathbf{r}_b}{s}, \qquad a_0 = \frac{1}{2+s}. \tag{15}$$

Noncollinearity gives $0 < s < 2$. Define

$$\begin{aligned} E'_a &= a_0(\mathbb{I} - \mathbf{r}_a \cdot \boldsymbol{\sigma}), \quad E'_b = a_0(\mathbb{I} - \mathbf{r}_b \cdot \boldsymbol{\sigma}), \\ E'_c &= a_0 s(\mathbb{I} + \mathbf{q} \cdot \boldsymbol{\sigma}), \quad E'_d = 0. \end{aligned} \tag{16}$$

Each effect is positive. Their scalar coefficients sum to $a_0(2+s) = 1$, and their Pauli coefficients sum to $-a_0(\mathbf{r}_a + \mathbf{r}_b) + a_0 s \mathbf{q} = 0$. Thus they sum to $\mathbb{I}$. Retain $\rho_a^\star, \rho_b^\star$, take $\rho'_c = (\mathbb{I} - \mathbf{q} \cdot \boldsymbol{\sigma})/2$, and choose any density operator for d. Then

$$\mathrm{Tr}(\rho'_i E'_i) = 0 \quad \text{for every } i. \tag{17}$$

If $A = \{a\}$, choose an inactive c and use instead

$$E'_a = \frac{\mathbb{I} - \mathbf{r}_a \cdot \boldsymbol{\sigma}}{2}, \qquad E'_c = \frac{\mathbb{I} + \mathbf{r}_a \cdot \boldsymbol{\sigma}}{2}, \tag{18}$$

with the other two effects zero, $\rho'_a = \rho_a^\star$ and $\rho'_c = (\mathbb{I} - \mathbf{r}_a \cdot \boldsymbol{\sigma})/2$. The remaining preparations are arbitrary. When A is empty, the same projective construction uses any two distinct labels

and any unit vector in place of $\mathbf{r}_a$. These targets are positive and complete, and Eq. (17) holds in both cases.

Keep all auxiliary effects fixed. Active preparation contributions are unchanged, and inactive auxiliary contributions remain state-independent. All scalar biases and row-shift constants are unchanged, while every target penalty remains zero. Therefore $L' = L^\star$. Since the reference already attains the global maximum, the replacement does as well. It has at least one zero effect, so it is inequivalent to four nonzero reference effects under common unitary or antiunitary transformations, including any outcome permutation. A single deterministic alternative suffices to prevent self-testing. Allowing shared randomness cannot eliminate that alternative; a linear score has the same maximum over strategies and their convex hull. This proves Proposition 1. □

4 Consequences and limitations

For exactly four matched pure kernel preparations, target self-testing by a globally optimal linear witness requires an auxiliary Bloch span of dimension three. At least three binary settings are therefore necessary in this reference class, although three settings need not suffice for arbitrary choices of the auxiliaries. The statement covers the entire coefficient array of the complete table and does not require independent certification of the preparations. The supporting coefficient form, followed by a physically admissible competing maximizer, extends the preparation-only span obstruction.

The replacement preserves the witness score, not the full probability table. In particular, no operational equivalence of the two realizations is asserted for measurements or statistics that are not evaluated through the chosen linear score. The matched-kernel hypothesis is used both in the rank-one dual slack and in preparation alignment. Self-tests with other reference preparations lie outside the result. Additional preparation inputs, nonlinear criteria, entanglement-assisted or sequential models, necessity of four preparations, and a noise-robust tradeoff are also outside its scope. The broader conjecture about resource minimality [2] consequently remains unresolved by this argument.

The derivation is exact, without a perturbative order or numerical tolerance. The accompanying symbolic checks reconstruct the full complementary-slackness solution space and verify positivity, completeness, diagonal exclusions and score equality for all eleven possible active-label sets on tetrahedral and unequal-weight extremal references. These finite checks support the algebra; the universal quantifiers follow from the analytic proof. No finite set of examples is used to infer the general obstruction.

Statements and declarations

Data and code availability. The exact symbolic verification code, its model definitions and regeneration instructions accompany this local draft in the reproducibility materials. No experimental data were generated. A public repository location has not been assigned.

Use of artificial intelligence. OpenAI Codex language-model tools were used extensively for literature discovery, proposed arguments, symbolic-code generation, algebraic checks, scientific auditing in a separate AI context, and manuscript drafting. The symbolic checks use exact arithmetic. AI checks do not constitute human scientific validation or external peer review. Human scientific review, authorship confirmation and approval of the final text remain pending for this local draft.

Author declarations. Author details, contributions, funding and competing-interest statements require confirmation by the named author before any submission.

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